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A Late Cretaceous pole for the Pacific plate: implications for apparent and true polar wander and the drift of hotspots

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Abstract

We present a Late Cretaceous (81 Ma) pole position for the Pacific plate derived from paleomagnetic analyses of basalt samples from Detroit Seamount (of the Hawaiian–Emperor seamounts) that were oriented using Brunhes-age overprints. This pole is at much higher latitudes than the previously published Late Cretaceous pole positions based on the modeling of magnetic anomalies observed during marine surveys over seamounts. Our new pole suggests that the Pacific plate would have moved rapidly between 95 and 81 Ma at speeds as high as 19.8 (− 10.8/ + 11.2) cm/year. The Pacific plate at this time was smaller than the present-day plate and had a substantial subducting boundary. The high-velocity estimates are comparable with those of other paleoplates having similar characteristics. Therefore, plate tectonic driving forces can explain the motion and there is no need to invoke true polar wander. Decreases in mantle drag associated with vigorous Late Cretaceous volcanism in the Pacific, however, may have contributed to the rapid plate speed. The new pole position, together with other reliable paleomagnetic indicators of Pacific apparent polar wander, further supports the notion of drift of the Hawaiian hotspot during the Late Cretaceous.
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Keywords: Pacific plate; Apparent polar wander; True polar wander; Hotspot

1. Introduction

The Pacific apparent polar wander (APW) path is marked by a standstill during the Early to Mid-Cretaceous, based on the analyses of basalt colatitudes from several Deep Sea Drilling Program (DSDP) and Ocean Drilling Program (ODP) sites (Tarduno and Sager, 1995). Until recently, it was thought that this standstill was followed by rapid latitudinal motion (~ 16 cm/year) of the Pacific plate between 81 and 72 Ma (Gordon, 1983; Sager and Pringle, 1988) (Fig. 1). This estimate was based primarily on Pacific poles determined by the modeling of seamount magnetic anomalies. However, colatitudes derived from standard paleomagnetic analyses of 81 Ma basalt recovered from drilling on Detroit Seamount (of the Hawaiian–Emperor seamounts) plot much further north when compared to these data (Tarduno and Cottrell, 1997). Paleomagnetic data from Early Campanian basalt recovered from drilling on Wodejebato Guyot (Tarduno and Gee, 1995) support the results from Detroit Seamount. Together, these data suggest a high rate of latitudinal motion similar to that suggested by the

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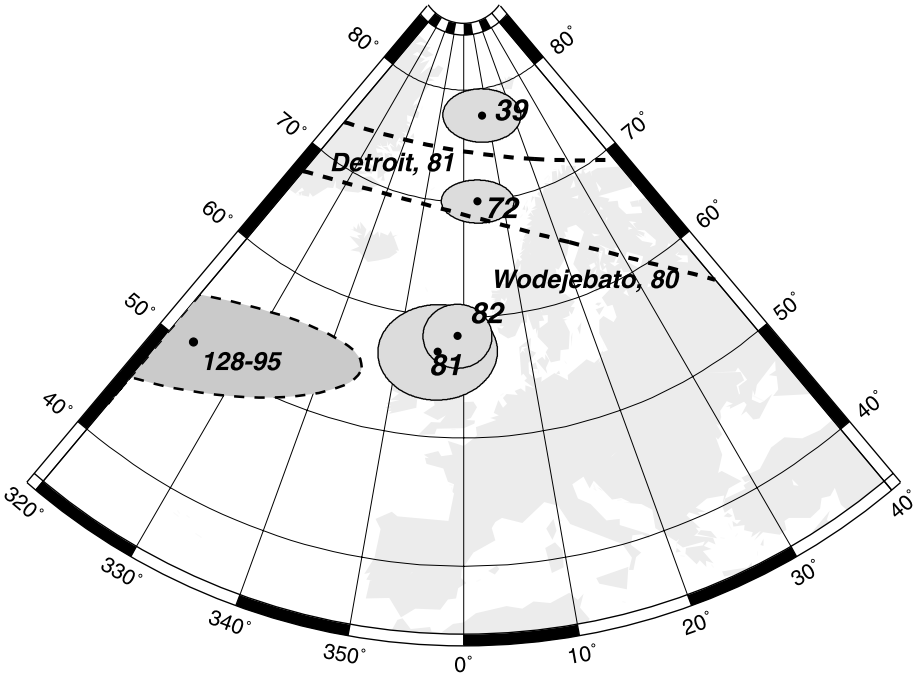


Fig. 1. Stereographic projection of data used to constrain Pacific apparent polar motion. Light ellipses show pole positions that include data derived from modeling of seamount magnetic anomalies (Gordon, 1983; Sager and Pringle, 1988). Darker ellipse with dashed outline shows pole constrained by paleomagnetic analyses of Mid-Cretaceous basalts recovered by ocean drilling (Tarduno and Sager, 1995). Heavy dashed lines show colatitudes derived from paleomagnetic analyses of Late Cretaceous basalts recovered by ocean drilling (Tarduno and Gee, 1995; Tarduno and Cottrell, 1997). The latter results are incompatible with Late Cretaceous pole positions based on the modeling of seamount magnetic anomalies (Cottrell and Tarduno, 2000).

poles based on seamount anomaly modeling (SAM), but at an earlier time period (95–81 Ma rather than 81–72 Ma). Therefore, there is a sharp difference in the timing of rapid Pacific APW that depends on the data sets used.

It is important to emphasize the differences in these data. SAM poles are based on the data collected during marine geophysical surveys over seamounts. The recovered data contain numerous spurious magnetizations that can obscure the primary (remanent) magnetization that might otherwise be useful for backtracking plate motion. These include magnetizations induced by the modern magnetic field of the Earth and viscous magnetizations acquired during the present Brunhes normal polarity chron (Gee et al., 1989). In addition, inhomogeneities caused by a given seamount having been formed at different times, possibly during different polarities of the Earth's field, add errors. The total observed anomaly must be modeled to derive a pole. The large uncertainties in the modeled pole positions have been addressed by Parker (1991).

In contrast, ocean drilling of seamounts can retrieve basalt that can be examined with standard paleomagnetic analyses, where induced magnetizations are excluded through measurements in magnetometers that are shielded from the Earth's magnetic field. Viscous magnetizations are removed through exhaustive laboratory demagnetization. A comparison of directions from lava flows allows one to gauge the amount of time represented in a given section.

Recently, Sager and Koppers (2000) modified Late Cretaceous SAM paleopoles to locations that agreed better with the colatitudes derived from Detroit Seamount and Wodejebato Guyot. These new Late Cretaceous SAM paleopoles cannot be considered independent measures of Pacific APW (Cottrell and Tarduno, 2000). Nevertheless, they do demonstrate that there is a general agreement that the

prior 81–82-Ma poles based on SAM data and the associated timing of rapid Pacific APW (i.e. 81–72 Ma) are no longer tenable.

The changes in the Pacific APW path forced by the new paleomagnetic data from the Detroit Seamount and Wodejebato Guyot, however, raise new questions concerning Late Cretaceous geodynamics. In particular, APW paths can also reflect true polar wander, the rotation of the entire solid Earth with respect to the spin axis, caused by changes in the distribution of mantle mass heterogeneities (Goldreich and Toomre, 1969). Can the relatively high rates of Pacific APW, approximately between 95 and 81 Ma, be reconciled by plate tectonic processes or is true polar wander required? We address these questions by a continued examination of paleomagnetic data from Detroit Seamount. In particular, several basalt core samples from Detroit Seamount have a magnetic overprint that was most likely acquired during the Brunhes normal polarity field (Tarduno and Cottrell, 1997). We use this overprint to provide azimuthal orientation for the paleomagnetic data allowing us to constrain a pole position rather than just a colatitude. We use these data to calculate the rates of plate motion that we further compare with plate tectonic factors important for controlling plate velocities.

2. Detroit Seamount (81 Ma) colatitude

Ocean drilling at Site 884E (ODP Leg 145) on Detroit Seamount recovered 87 m of basalt that were separated into 13 lithologic units based on phenocryst content and the presence of chilled margins (Rea et al., 1992). A total of 94 azimuthally unoriented samples were measured for characteristic remanent magnetization (ChRM) directions using detailed stepwise thermal demagnetization techniques. Magnetic hysteresis properties and unblocking temperatures for the basalts suggested a magnetic mineralogy of magnetite and low-titanium titanomagnetite (Tarduno and Cottrell, 1997). ChRM inclinations were separated into 10 inclination groups that were distinct from each other at the 95% confidence level. Angular dispersion, estimated from the inclination dispersion of the 10 inclination groups, is indistinguishable from that predicted for global volcanic data of comparable age (McFadden et al., 1991). The colatitude of Detroit Seamount [53.8 ($-7.2/+6.9$)], when plotted against the Pacific APW path, is distinct from the prior Late Cretaceous poles of Gordon (1983) and Sager and Pringle (1988) at the 99% confidence interval (Fig. 1).

3. Detroit Seamount (81 Ma) pole

Orthogonal vector plots of basalt samples analyzed from Detroit Seamount showed univectorial decay to the origin after the removal of a viscous overprint (Tarduno and Cottrell, 1997). In $\sim$10% of the samples, the overprint was well defined in the demagnetization data and a vector could be fit using principal component analysis (Table 1). The overprint inclinations are similar to those expected for the Detroit

Table 1
Samples used for Detroit pole determination

I Grp.	Sample	ChRM I (°)	ChRM D (°)	OP I (°)	OP D (°)	OP MAD (°)	Rot. D (°)
B	4r04075	−57.4	13.2	58.7	173	4.0	200.2
	5r02016	−47.8	73.9	64.1	232.8	10.3	201.1
C	5r04085	−53.5	224.7	66.8	70.1	6.4	154.6
	5r05012	−55.8	304.7	57.3	115.1	6.8	189.6
I	10r05066	−51.3	108.1	54.7	296.5	8.7	171.6
	10r06018	−61.2	6.3	56.3	186.5	5.6	179.8
	10r06055	−58	98.6	60.0	271.5	5.3	187.1
	10r06089	−57	14.9	60.8	195.3	11.7	179.6
J	10r06123	−54.5	348.1	54.1	173.5	10.3	174.6
	10r06127	−53.7	338.9	59.0	158.3	11.3	180.6

I Grp. refers to the preferred inclination grouping of basalt units from Tarduno and Cottrell (1997). ChRM (characteristic remanent magnetization) I, ChRM D, OP (overprint) I, OP D and OP MAD are the directions and dispersions determined through principal component analysis. Rot. D is the rotated declination value.

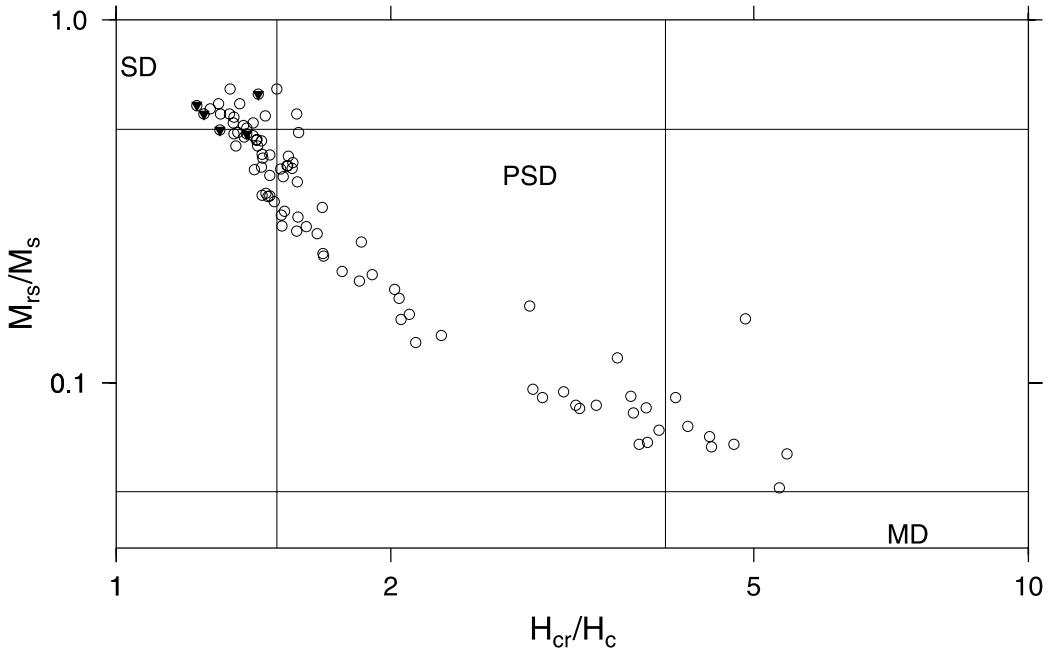


Fig. 2. Day plot (Day et al., 1977) of magnetic hysteresis data derived from basalt samples from Detroit Seamount. All magnetic hysteresis data were obtained with a Princeton Measurements alternating gradient force magnetometer at the University of Rochester. M_{rs} is the saturation remanence, M_s is the saturation magnetization, H_c is the coercivity and H_{cr} is the coercivity of remanence. Samples with magnetic hysteresis data used in pole position calculations (see text) are shown with solid inverted triangles. SD, single domain. PSD, pseudo-single domain. MD, multidomain.

Seamount site in a Bruhnes normal polarity epoch. The expected inclination for the Detroit Seamount site is 68.3° based on the geocentric axial dipole (GAD) hypothesis. The inclination only average of the overprint inclination directions (McFadden and Reid, 1982), is only slightly shallower [59.7° (−5.9°/+5.7°), α_{95}, $n=4$].

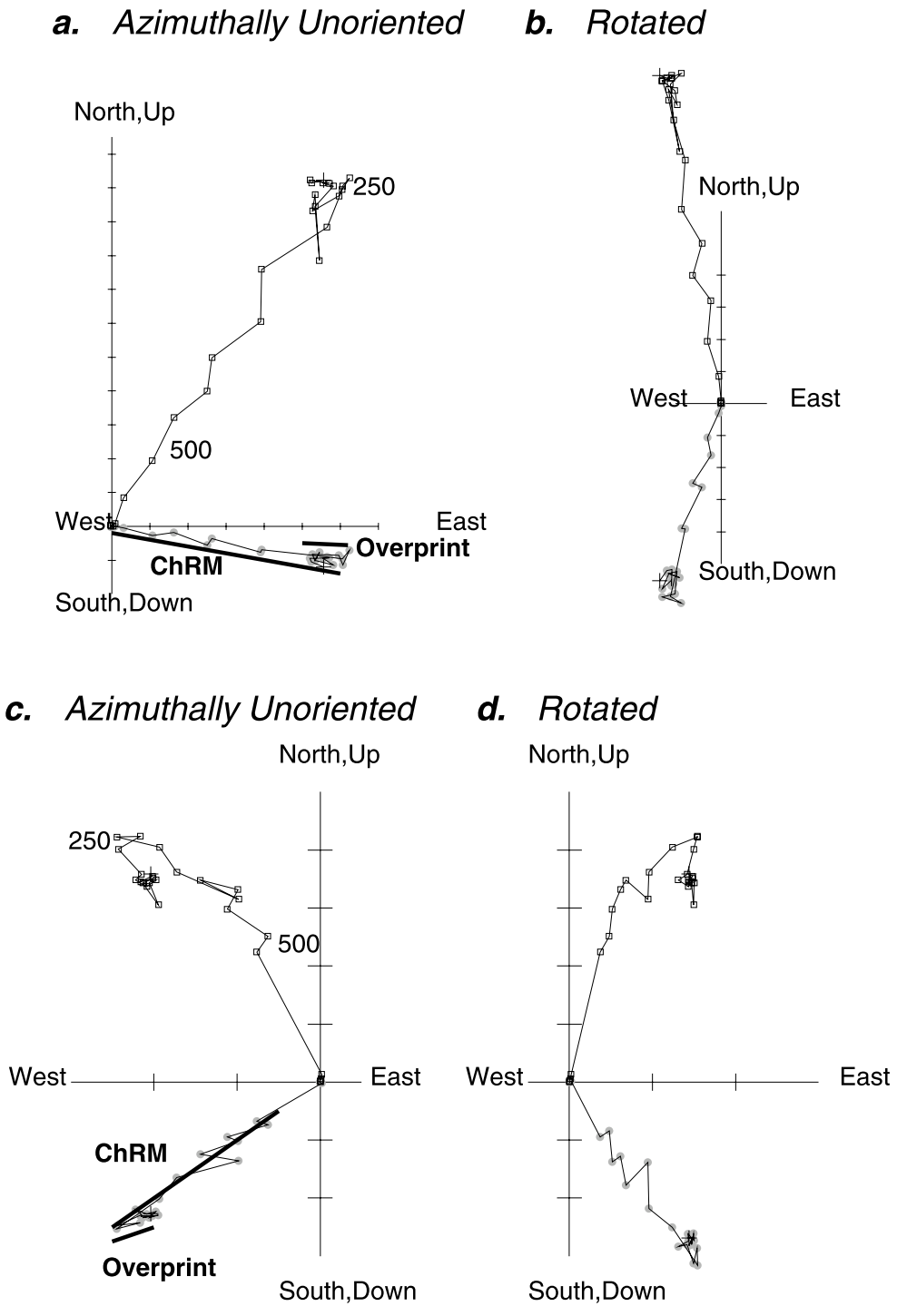
Bruhnes-age overprints can be used to define the rotation required to orient azimuthally unoriented samples, thereby defining true characteristic declination directions. The use of overprints in this way relies on the acquisition of a viscous remanent magnetization (VRM) and the GAD hypothesis. The intensity of VRM acquired will be dependent on the grain volume, temperature, time and field intensity (Butler, 1992). The time-averaged geomagnetic field is that of a geocentric dipole where the geomagnetic pole coincides with the spin axis (declination = 0°). Given sufficient time, some magnetic grains have enough thermal energy to overcome the anisotropy energy boundaries and will align with the ambient (Earth's) field acquiring a VRM (Dunlop and Özdemir, 1997).

For an assemblage of single domain grains, the VRM component can be demagnetized according to the following formula (Dunlop and Özdemir, 1997):

$$\frac{T}{\beta^2(T)} = \frac{\ln(2t/\tau_0)}{\ln(2t_c/\tau_0)} T_0, \quad (1)$$

where T is the temperature required to thermally demagnetize the VRM component, t is the time of acquisition, τ_0 is the atomic reorganization time, t_c is a

Fig. 3. Thermal demagnetization of sample showing a relatively large overprint removed at low unblocking temperatures. All samples were measured using a 2G Enterprises superconducting magnetometer at the University of Rochester. The magnetization overprint in the azimuthally unoriented sample (a, c) was assumed to record the Brunhes axial geocentric dipole field and was used as a reference to determine the true characteristic remanent magnetization (ChRM) declination. The ChRM declination reoriented following this method is shown in (b, d).


a. Azimuthally Unoriented
b. Rotated
North,Up
250
500
West
East
ChRM
Overprint
South,Down
North,Up
West
East
South,Down
c. Azimuthally Unoriented
d. Rotated
North,Up
250
500
West
East
ChRM
Overprint
South,Down
North,Up
West
East
South,Down

characteristic time determined from the heating or cooling rate through the blocking temperature, T_0 is the temperature at which the VRM was acquired and $\beta(T)$ is an estimate of the normalized saturation magnetization of single domain grains as a function of temperature (Dunlop and Özdemir, 1997):

$$\beta(T) \equiv \frac{M_s(T)}{M_{s0}} = \left(\frac{T_c - T}{T_c - T_0}\right)^{0.43}, \quad (2)$$

where T_c is the Curie temperature, $M_s(T)$ is the saturation magnetization at temperature T and M_{s0} is the saturation magnetization at room temperature.

We can compare a temperature range of T estimated by Eqs. (1) and (2) with the temperature range determined from component analysis of the overprints in the orthogonal vector plots. Magnetic hysteresis properties of some of the samples showing a clear overprint direction plot near the single domain field of a 'Day-plot' (Day et al., 1977) (Fig. 2) suggest that the VRM theory we are considering here is valid. Assuming a t of 0.78 million years (the length of the present Brunhes normal polarity interval), t_c is 100 s, τ_ϕ is 10^{-9} s, $T_\phi = 25–30$ °C and Curie temperatures of 500–580 °C (approximated by the unblocking characteristics of the basalts), T can range from ~150 to 200 °C. The temperature range needed to demagnetize the overprint (200–250 °C) is close to the upper part of this range supporting a viscous origin.

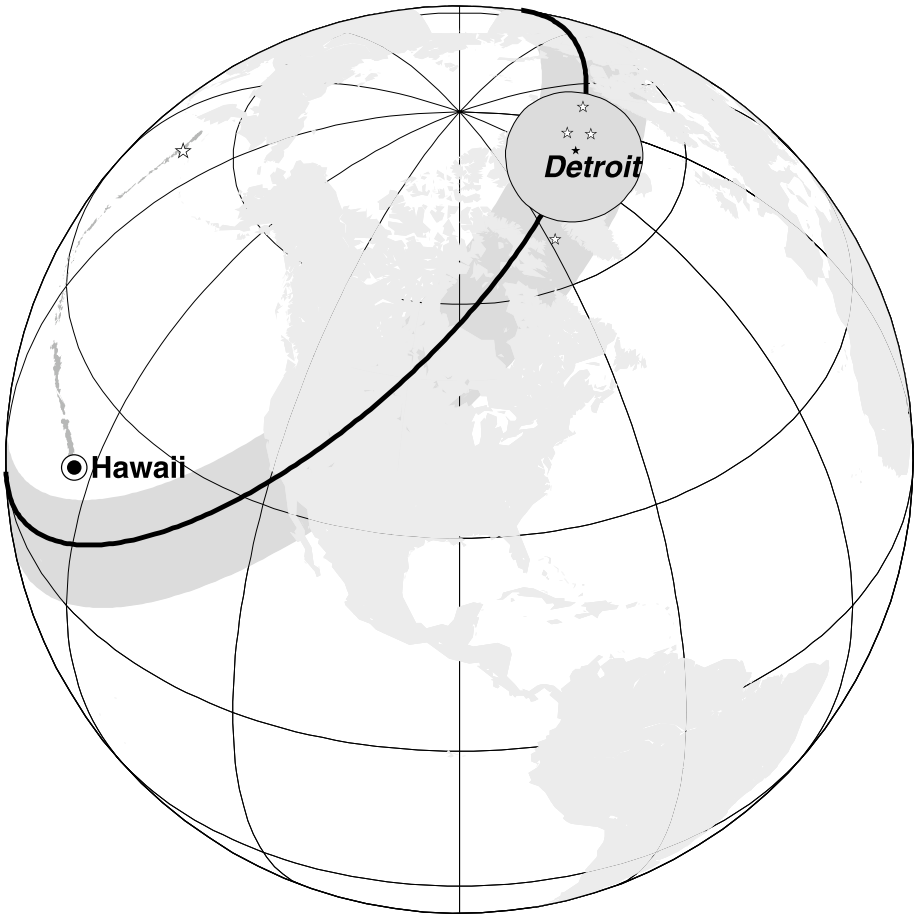


Fig. 4. Orthographic projection of colatitude and 95% confidence interval derived from paleomagnetic inclination data from Detroit Seamount (Tarduno and Cottrell, 1997). Also shown is the mean pole position and 95% confidence interval based on azimuthally oriented data (see text). White stars represent virtual geomagnetic poles used to determine the mean pole position.

The angle required to rotate the azimuthally unoriented overprint to the GAD direction (i.e. declination = 0°) was applied to the azimuthally unoriented ChRM declination direction to obtain a true (oriented) declination direction (Fig. 3). Overprint directions were fit by principle component analysis to 12 samples representing six inclination groups from Tarduno and Cottrell's (1997) preferred 10-inclination group model. However, two inclination groups have only one sample each and were omitted from further analysis. Inclination groups with more than one sample were averaged together to determine a group declination and inclination. These directions were next transformed into pole space (Irving, 1964) and confidence limits were calculated based on Fisher statistics (Butler, 1992). The pole position for the azimuthally oriented paleomagnetic data from Detroit Seamount (74.3° N, 341.1° E, $n = 4$) and its confidence interval ($dm_{95} = 12.1°$; $dp_{95} = 8.6°$ at an azimuth of 87.6°) is shown in Fig. 4.

4. Discussion

The new pole position derived from Detroit Seamount, together with a previously published Mid-Cretaceous Pacific standstill pole (Tarduno and Sager, 1995), allows us to estimate the full rate of Pacific plate motion between the mid-Cretaceous (95 Ma) and the Late Cretaceous (81 Ma). This rate is 19.8 ($-10.8/+11.2$) cm/year (95% confidence interval quoted). We note, however, that the age of the Mid-Cretaceous Pacific standstill is not well constrained. Therefore, our calculated plate speed could be an overestimate if apparent polar wander rates increased prior to 95 Ma. For example, if we use a midpoint age of the Mid-Cretaceous pole (111.5 Ma), the calculated plate velocity is similar to that of the present-day Pacific plate [9.5 ($-5.0/+5.1$) cm/year]. Because the relevant question is whether mid-Late Cretaceous Pacific APW is too rapid to be explained by plate tectonic driving mechanisms, we will retain the higher velocity estimates in the following discussion.

The 19.8 ($-10.8/+11.2$) cm/year plate velocity is higher than that of the present-day Pacific plate (Minster and Jordan, 1978) but not unprecedented when the speeds of other past plates are considered. For example, the Indian plate may have moved as fast as 14 cm/year during the Paleocene to Early Eocene (Schult and Gordon, 1984). While slab-pull is generally recognized as the dominant plate driving force (Forsyth and Uyeda, 1975), the areal extent of a plate may also be important as mantle drag is proportional to the plate area. Relatively small oceanic plates similar in size to the paleo-Indian plate (e.g. Kula and Izanagi) subducting old crust could have had even higher plate velocities since they lacked a continental drag force (Tarduno et al., 1990). Based on the reconstructions of Engebretson et al. (1985), the Kula plate may have been twice the size of the paleo-Indian plate though both appear to have had similar plate velocities (~ 14 cm/year). The Izanagi plate was slightly larger, and is estimated to have reached velocities as high as ~ 25 cm/year (Engebretson et al., 1985; Tarduno et al., 1990).

Absolute locations of magnetic isochrons were assigned by approximating the Pacific apparent polar wander path by two tracks (0–81 and 81–95 Ma), utilizing our new pole from Detroit Seamount and the Mid-Cretaceous standstill pole (Tarduno and Sager, 1995). Each track was fit by a paleomagnetic Euler pole ($\lambda = 3°$ N, $\phi = 110°$ E and $\lambda = 12°$ N, $\phi = -133°$ E) and these poles were used to rotate magnetic isochrons back in time. Clearly, the Pacific APW path is underdetermined and the approximation we use here is meant to provide only a first-order picture of plate history. Nevertheless, this approach allows us to view this history independent of the fixed hotspot hypothesis that is now seriously in question (Tarduno and Gee, 1995; Tarduno and Cottrell, 1997).

Schematic representations of the configuration of plates in the Cretaceous Pacific Basin, which draw on the relative plate motion scenarios proposed by Engebretson et al. (1985) and Hilde et al. (1977), are shown in Fig. 5. Positions of subduction zones and transform boundaries are based on relative motion vectors from Engebretson et al. (1985) and Hilde et al. (1977). Model A represents a possible plate configuration near the beginning of the Cretaceous Normal Polarity Superchron. The Farallon plate dominates the Pacific basin in the east while the Izanagi plate is subducting beneath the Eurasian and Farallon plates. Model B represents a possible configuration between 110 and 95 Ma. The Phoenix plate has been transferred to the Pacific plate increasing its surface area by at least 20% (the boundary

between the Phoenix/Pacific and Austral–Antarctic plates is not well constrained).

Model C represents our preferred plate configuration for the Late Cretaceous period of fast apparent polar wander. Near the end of the Cretaceous Normal Polarity Superchron, the Izanagi plate was captured by the Pacific plate. This was followed by the formation of the Kula plate that broke from the Farallon plate at approximately 84 Ma (Wood and Davies, 1982). Ridge boundaries probably existed between the Pacific and Bellinghousen (Hilde et al., 1977; Engebretson et al., 1984) and Austral–Antarctic plates (Hilde et al., 1977). Subduction boundaries are assumed for most of the northern Pacific plate based on relative plate motion vectors with respect to Eurasia presented by Engebretson et al. (1985).

Using these plate reconstructions, we can now compare our estimate of Cretaceous Pacific plate motion with those of other Pacific basin plates, and examine all these motions with respect to the percentage of the plate connected to a subducting slab (Fig. 6). The Pacific plate of Model C has a substantially

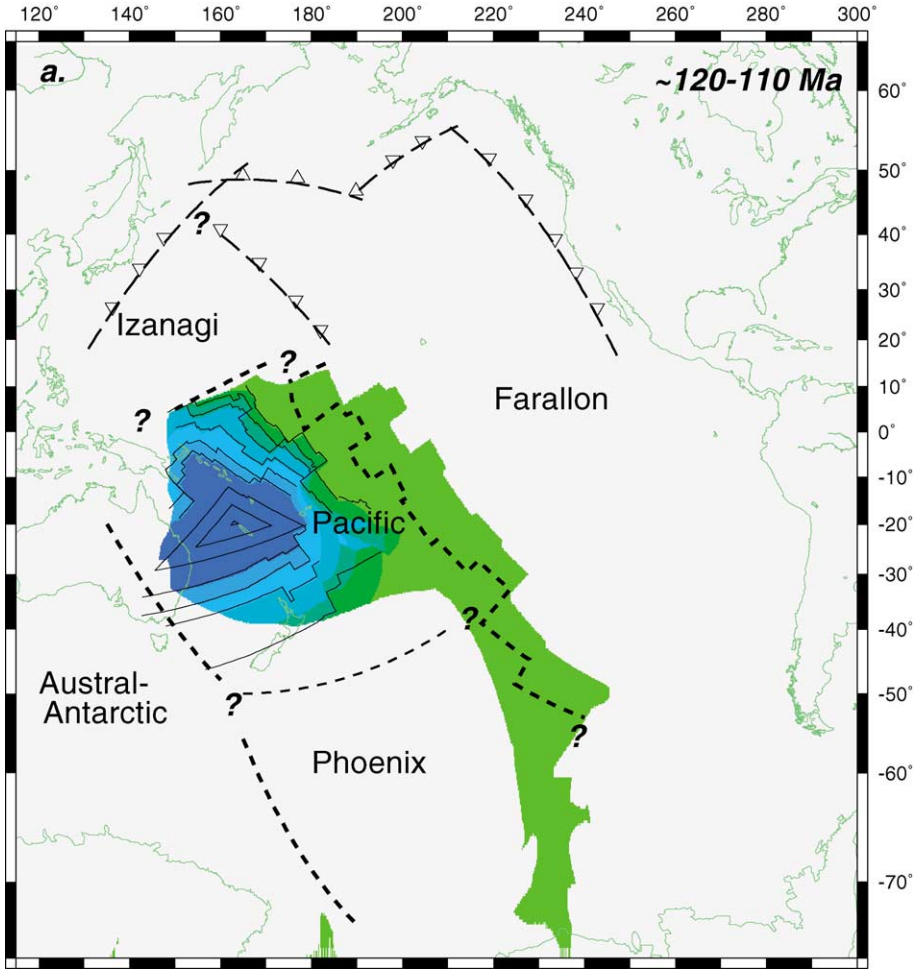


Fig. 5. (a) Hypothetical plate configuration for the Pacific Basin between 120 and 110 Ma following the scenarios of Engebretson et al. (1985) and Hilde et al. (1977). Magnetic isochrons were rotated to their position at the beginning of the Cretaceous Normal Polarity Superchron using paleomagnetic Euler poles fit to Pacific apparent polar wander constrained by paleomagnetic data from basalts recovered by ocean drilling (see text). (b, c) Hypothetical plate configurations for 110–95 and 90–81 Ma following methods in (a). Large closed circle in (c) represents the reconstructed position of the Hawaiian hotspot at 81 Ma.

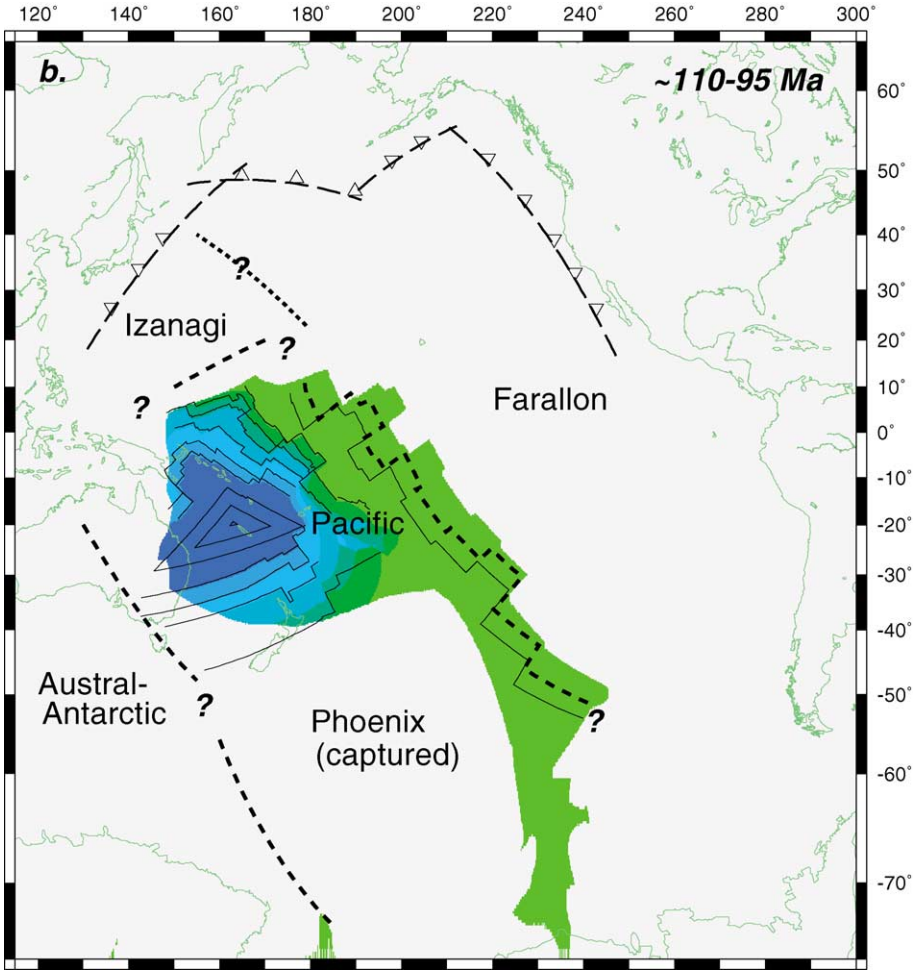


Fig. 5 (*continued*).

larger subduction zone boundary than Models A and B. This observation agrees with the available paleomagnetic data: Models A and B represent the Cretaceous Normal Polarity Superchron where the Pacific APW path shows a standstill (Tarduno and Sager, 1995), while Model C represents the time of rapid APW presented here.

Past plate motion has also been inferred for the Farallon plate between 101 and 89 Ma (14–30 cm/year; Alvarez et al., 1980; Tarduno et al., 1990; Tarduno and Myers, 1994). We note that the Farallon and Pacific plates had similar plate areas during these times of fast motion.

Another factor important for controlling plate velocity is mantle viscosity that can decrease with increased plume activity (Morgan and Smith, 1992). As a result, mantle drag can be reduced assisting rapid plate motions. Interestingly, the Mid- to Late Cretaceous was a time marked by tremendous oceanic mantle plume volcanism (Tarduno et al., 1998). In particular, the Caribbean plateau formed mainly between 91 and 88 Ma (Sinton et al., 1998); the large mantle plume, thought to have been responsible for its formation, could have been a factor in decreasing mantle drag forces in the eastern Pacific during the Late Cretaceous.

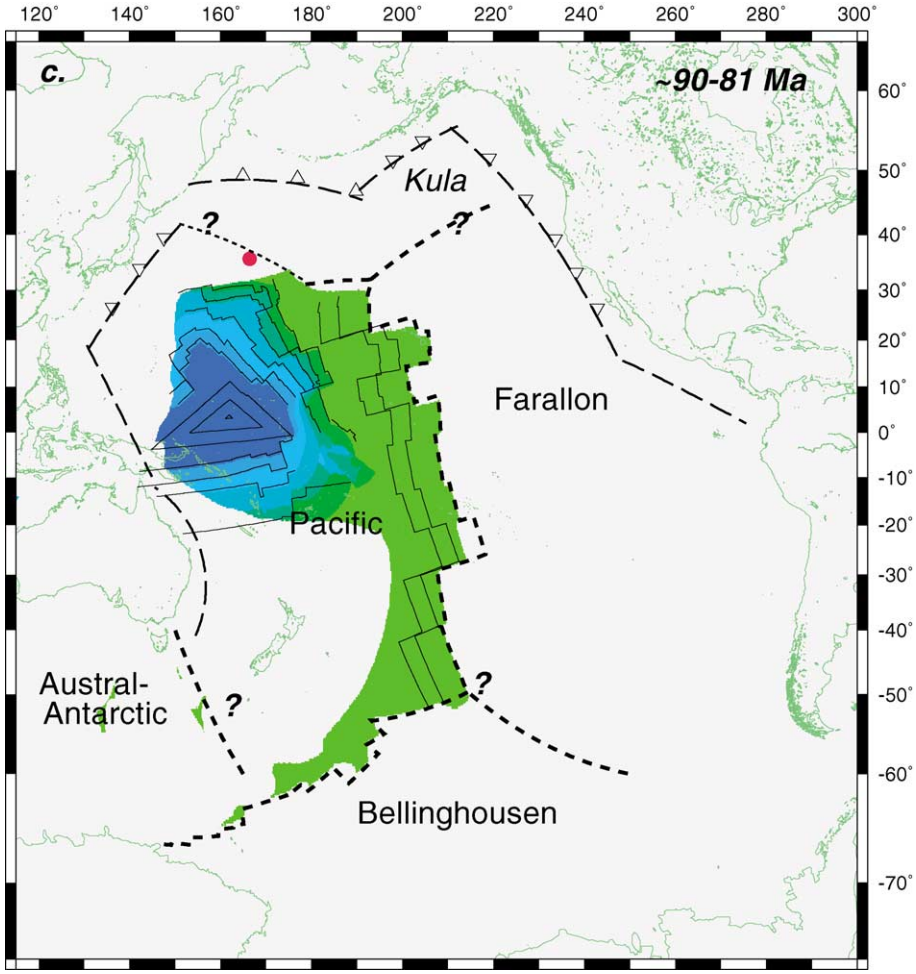


Fig. 5 (*continued*).

After approximately 81 Ma, the Pacific plate appears to have experienced much lower velocities. This conclusion is supported by basalt colatitude data from Suiko Seamount (Kono, 1980) and the results of the analyses of marine magnetic anomaly skewness (Fig. 7). The latter can provide important additional information of the apparent polar motion of ocean plates, but they suffer from relatively large uncertainties associated with factors related to how oceanic crust acquires a magnetization and/or the behavior of the paleomagnetic field during crustal formation. These uncertainties are generally summarized in an "anomalous skewness" factor that can be approximated and corrected for if data are available for both sides of a ridge segment (Cande, 1976). However, this is not the case in most of the Cretaceous plates considered here. Nevertheless, techniques have been developed to estimate anomalous skewness from single plates (Petronotis and Gordon, 1999).

Anomaly skewness poles for chron 32 (Petronotis and Gordon, 1999) and chron 33n (Vasas et al., 1994) are very close to each other, suggesting little apparent polar wander between 79 and 73 Ma (Fig. 7). Both of these poles lie close to the new 81-Ma pole position derived from Detroit Seamount. A 65-Ma pole based on anomaly skewness (Acton and Gordon, 1991) also falls within the colatitude distribution derived from Detroit Seamount.

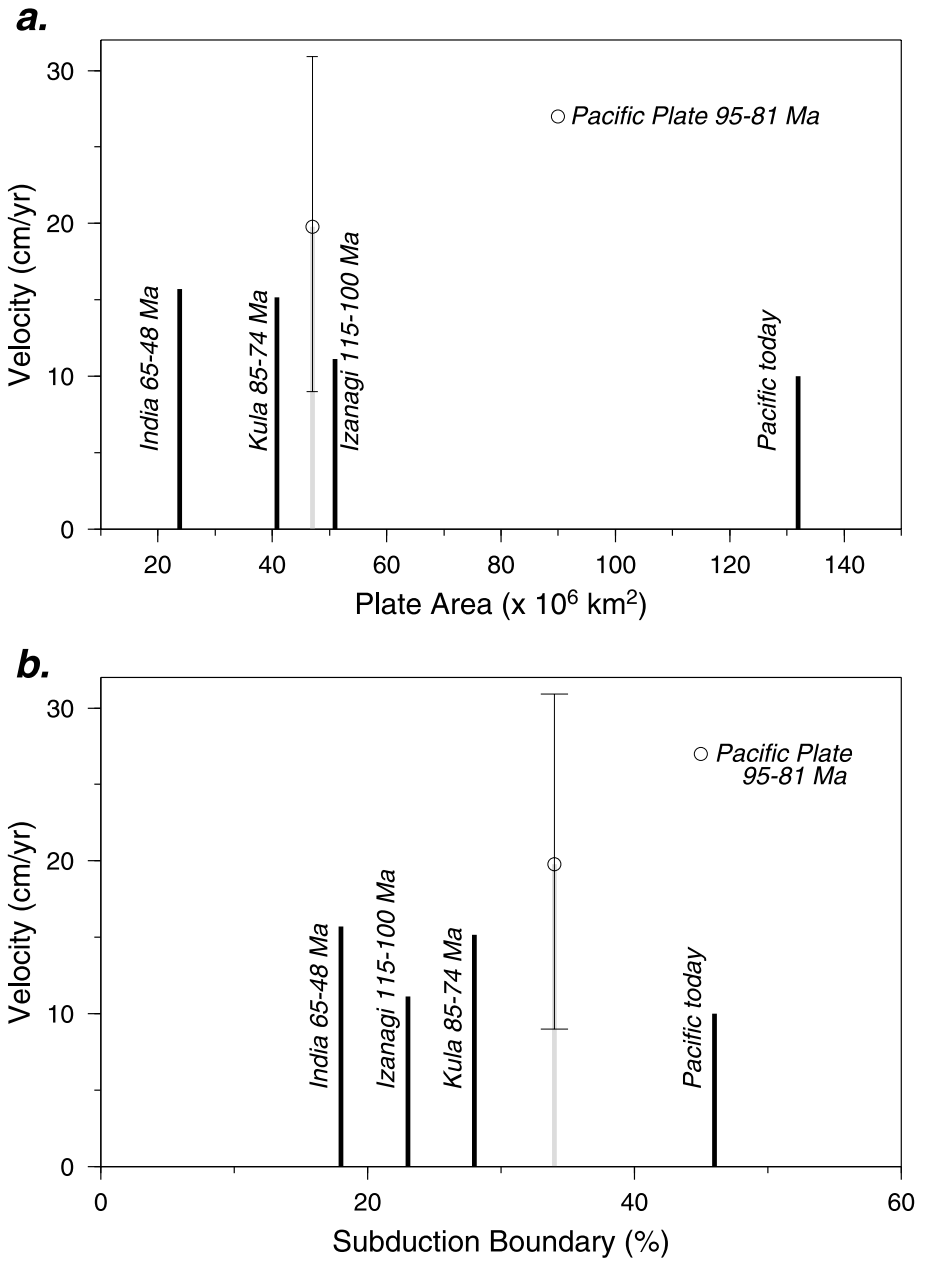


Fig. 6. (a) Comparative plate velocities for the paleo-India (Schult and Gordon, 1984), Kula, Izanagi (Engebretson et al., 1985) and Pacific (this study) plates and the modern Pacific plate. These velocities are plotted versus estimated plate areas. (b) Velocities shown versus plate subduction boundary percentages.

Tarduno and Cottrell (1997) suggested that the discrepancy between the paleolatitude of Detroit Seamount and the present-day location of the Hawaiian hotspot was due to the motion of the Hawaiian hotspot between 81 and 65 Ma. If the Cretaceous anomaly skewness poles discussed above are accurate recorders of APW (Fig. 7), the Emperor seamounts, rather than showing a progression

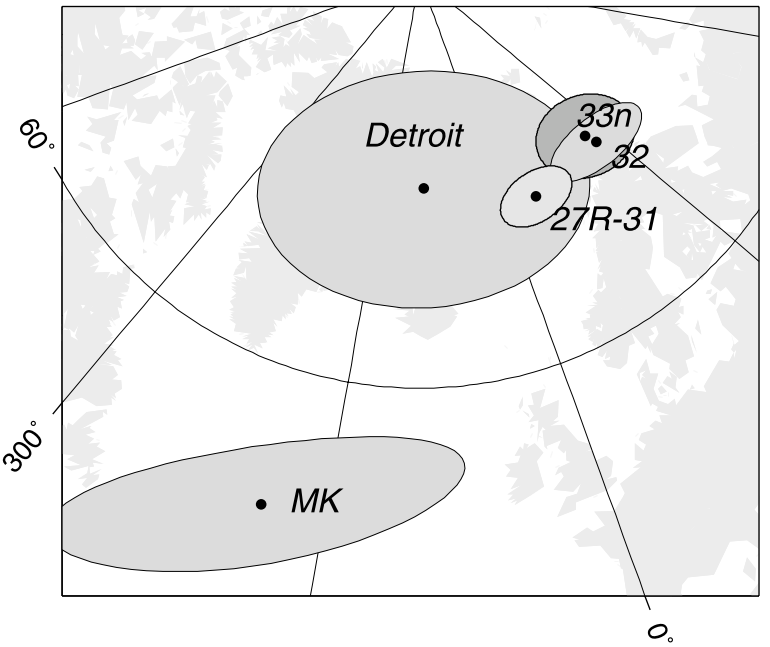


Fig. 7. Comparison of pole positions for the Cretaceous Pacific plate based on basalt colatitudes (128–95 Ma, Tarduno and Sager, 1995; 81 Ma, Detroit Seamount, this work) and marine magnetic anomaly skewness (chron 33n, Vasas et al., 1994; chron 32, Petronotis and Gordon, 1999; and chrons 27R-31, Acton and Gordon, 1991).

of plate motion during this time period, shows evidence of a hotspot moving beneath the Pacific plate.

5. Conclusions

The availability of basalt paleomagnetic data from Detroit Seamount and other sites has allowed us to revise considerably the Cretaceous apparent polar wander path of the Pacific plate. A new interval of rapid apparent polar wander has been identified during the Late Cretaceous (~95–81 Ma). At this time, the Pacific plate, which was much smaller than the present plate but still had a considerable subducting perimeter, has reached speeds as high as 19.8 (−10.8/+11.2) cm/year. These velocities are compatible with plate tectonic driving forces and there is no need to invoke true polar wander. However, decreases in mantle drag associated with vigorous Late Cretaceous mantle plume activity may have contributed to these rapid motions. The new apparent polar wander path further supports the notion of considerable motion of the Hawaiian hotspot during the Late Cretaceous.

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